

VibeAthon6.0 (Vibe Coding Hackathon) – 2K26

Problem Statement (PS): Smart Restaurant Management System

Duration: 3 Days

Mode: Online | Individual / Team

Date: 25th July 2026 – 27th July 2026

OBJECTIVE

Build and deploy a full-stack SaaS platform that improves restaurant operations by solving real-world challenges faced by both customers and restaurant management.

Rather than replicating existing food delivery platforms, participants are expected to identify operational problems, research possible solutions, and build an innovative product that enhances the overall dining experience.

CHALLENGE OVERVIEW

Restaurants continue to rely on manual processes, resulting in operational inefficiencies and poor customer experiences.

Common challenges include:

- Customers waiting to know whether dishes are available.
- Limited visibility into menu items and restaurant services.
- Long waiting times for tables and orders.
- Delayed communication between customers, staff, and kitchen.
- Manual order, billing, and inventory management.
- Inefficient staff coordination.
- Lack of operational insights and business analytics.

Participants should first understand these challenges and then design technology-driven solutions that improve restaurant operations from end to end.

USER STORIES

Bronze Level – User Experience

User Story 1

Design a modern and intuitive interface for both customers and restaurant management.

The solution should clearly demonstrate how technology improves the restaurant experience.

Silver Level – Authentication & Digital Operations

User Story 2

Implement secure authentication using:

- Email & Password with OTP (or equivalent verification)
- Google OAuth

Support appropriate role-based access where required.

User Story 3

Digitize core restaurant workflows based on your proposed solution.

Examples include:

- Digital menu
- Live item availability
- Smart reservations
- Order management
- Queue management
- Billing
- Customer notifications

Participants are free to decide which workflows best solve their identified problems.

Gold Level – Restaurant Management

User Story 4

Develop a management dashboard that enables restaurant staff to efficiently manage daily operations.

Examples include:

- Orders
- Tables
- Inventory
- Staff
- Customers
- Sales
- Analytics

The dashboard should reduce manual effort and improve operational efficiency.

Platinum Level – Intelligent Operations

User Story 5

Integrate intelligent features that improve decision-making and customer experience.

Examples include:

- Personalized recommendations
- Inventory prediction
- Demand forecasting
- Smart notifications
- Operational insights
- AI-powered assistance

Participants are encouraged to research and implement innovative capabilities beyond the listed examples.

Ranking System

- **Bronze** – User Story 1
 - **Silver** – User Stories 1–3
 - **Gold** – User Stories 1–4
 - **Platinum** – User Stories 1–5
 - **Bonus** – Additional innovative features
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Suggested Technical Stack

- **Frontend:** React.js / Next.js (or equivalent)
 - **Backend:** Node.js with Express.js
 - **Database:** MongoDB / PostgreSQL / Supabase
 - **Authentication:** Email/Password + Google OAuth
 - **AI (Optional):** Gemini API or equivalent
 - **Deployment:** Vercel, Netlify, Render, Railway, etc.
 - **Version Control:** GitHub with meaningful commit history
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Submission Requirements

1. Hosted application must be live and publicly accessible
2. GitHub repository must be in public
3. README including:
 - Team Name
 - Tech Stack
 - User Stories Completed
 - AI Usage (if applicable)
 - Hosted Application Link
4. Must use Given PPT and Submit in PDF Format (will be given at day 3)

Judging Criteria

- Functionality
- User Experience
- Innovation
- Problem Solving
- Code Quality
- Scalability
- Deployment

Important Notes

- Do not build a clone of an existing restaurant application.
- Focus on solving operational challenges through technology.
- Research real-world restaurant workflows before designing your solution.
- Participants are encouraged to extend the platform with additional features that strengthen their product vision.